



Year 11 Mathematics

Semester 2 Examination

September 2010

Time Allowed: 2 Hours

Instructions:

- Start each question on a new page
- Show all necessary working
- Silent, non-programmable calculators may be used
- Pencil or liquid paper should not be used
- Answer all questions
- Marks may not be awarded for partial or incomplete answers

+ Solutions
Caringbah 2010 Y11 2H Yearly
Q & S
2H Y11 Yearly
2010 Q & S

✓ Basic
✓ Calculus
✓ Trig
✓ Geometry
✓ Coordinate
✓ Functions

QUESTION 1: (13 marks)

Marks

i) State the domain and range of $y = 9 - x^2$

2

ii) Simplify: $\frac{x^2 + 3x + 2}{x^3 + 8}$

2

iii) Solve: $5x^2 + 7x - 12 = 0$

2

iv) Differentiate

(a) $y = 2x^3 - 12x + 7$

1

(b) $y = \frac{3}{x^4}$

2

v) Simplify: $\frac{a+2}{5} - \frac{a-1}{3}$

2

vi) Factorise fully: $x^3 - x + x^2y - y$

2

QUESTION 2: (13 marks)**Marks**

i) If $\frac{5}{2\sqrt{2}+3} = a + b\sqrt{2}$, find the values of a and b

2

ii) Express $y^{\frac{-2}{3}}$ without a negative or fractional index

1

iii) Draw a neat sketch of the following functions:

(a) $y = \frac{-2}{x}$

2

(b) $y = \sqrt{4-x^2}$

2

iv) Find $\frac{dy}{dx}$ given $y = (5-2x)^3$

2

v) Given $\sin x = \frac{3}{7}$ and $\tan x < 0$, find the exact value of $\cos x$

2

vi) Evaluate: $\lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2}$

2**QUESTION 3: (13 marks)**

i) If $f(x) = \sqrt{4x-3}$, find the value of $f'(3)$

2

ii) Solve simultaneously: $\begin{cases} x-y=1 \\ xy=2 \end{cases}$

2

iii) Simplify: $\frac{2^n \times 4^{n+1}}{8^{n-2}}$

2

iv) Differentiate: $y = \frac{x^2}{3x+1}$

2

Question 3 continued over page

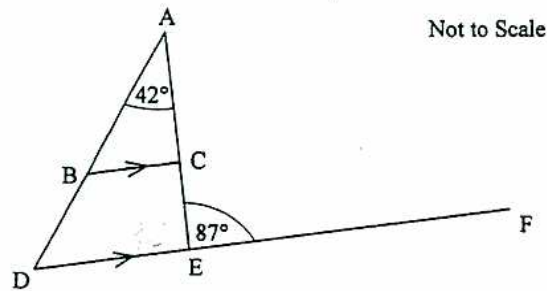
QUESTION 3: (cont)**Marks**

- v) In the diagram below, ADE is a triangle. B and C lie on AD and AE respectively such that BC is parallel to DE. Line DE is produced to F.

3

$$\angle AEF = 87^\circ \text{ and } \angle DAE = 42^\circ$$

Find the size of $\angle ABC$, giving reasons for your answer.



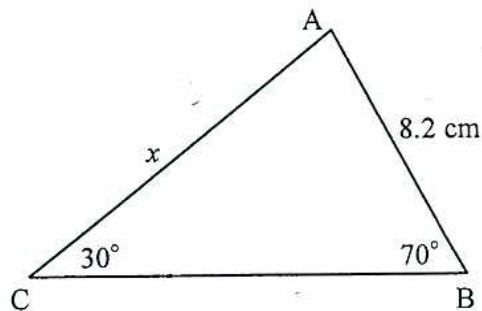
- vi) Given $f(x) = x^2 - 1$, is $f(a+b) = f(a) + f(b)$? Justify your answer.

2**QUESTION 4: (13 marks)**

- i) Find:

- (a) the length of AC, correct to 1 decimal place

- (b) the area of the triangle ABC, to the nearest cm^2 .



NB. Diagram not to scale

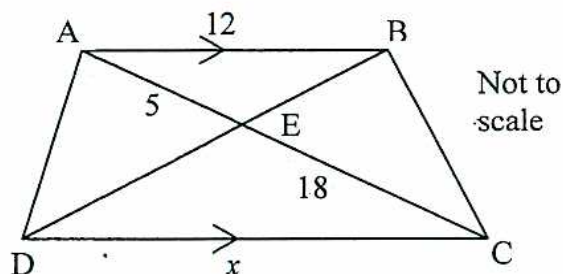
- ii) Solve: $|2x - 5| = 3x + 1$

- iii) Differentiate by first principles: $f(x) = x^2 - 2x$

Question 4 continued over page

QUESTION 4: (cont)**Marks**

iv)



(a) Copy the diagram into your book.

(b) Prove $\triangle ABE$ is similar to $\triangle CDE$.

3

(c) Find the length of x , giving reasons.

1**QUESTION 5: (13 marks)**

Given the points $A(1,4)$, $B(-2,-1)$ and $C(4,3)$,

(a) Find the length of the interval joining B to C

1

(b) Find M , the midpoint of the interval BC

1

(c) Find the equation of the circle, centre M , passing through both B and C

2

(d) Find the gradient of the line joining B to C

1

(e) Show that the equation of the line BC is $2x - 3y + 1 = 0$

1

(f) Find the perpendicular distance from A to the line BC

2

(g) Hence find the area of triangle ABC

1

(h) Find the equation of the line through B perpendicular to the line BC

2

(i) Find the co-ordinates of the point $D(x,y)$ such that A is the midpoint of BD

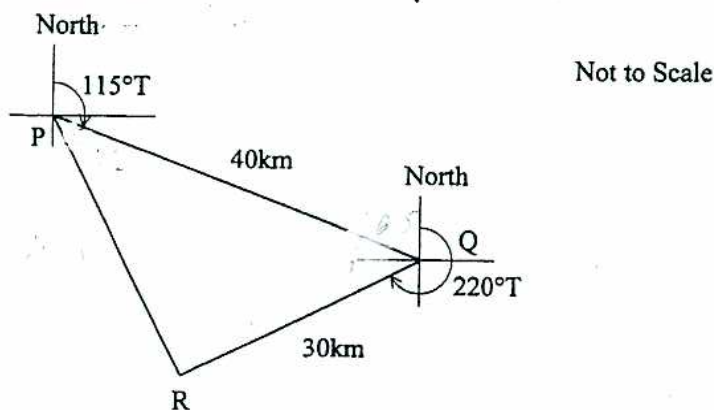
2

QUESTION 6: (13 marks)**Marks**

i) Prove: $\frac{1}{1-\sin x} + \frac{1}{1+\sin x} = 2\sec^2 x$

3

- ii) From P the bearing of a Lighthouse Q, 40 kilometres distant from P, is 115°T . From Q the bearing of a headland R, 30 kilometres from Q, is 220°T . This is illustrated in the diagram below.



- (a) Find the size of $\angle PQR$
- (b) Use the Cosine Rule to find the length of PR. Give your answer correct to 2 decimal places.
- (c) Find the bearing of R from P. Give your answer to the nearest whole degree.

1**1****2**

iii) For $y = \frac{6}{\sqrt{x}}$ find:

(a) $\frac{dy}{dx}$

2

(b) The gradient of the curve $y = \frac{6}{\sqrt{x}}$ at the point where $x = 9$

1

iv) A function is defined as: $f(x) = \begin{cases} 1 & \text{for } x < 0 \\ 2^x & \text{for } x \geq 0 \end{cases}$

- (a) Sketch the graph of this function for $-3 \leq x \leq 3$

2

(b) Find the value of $f(-2) + f(2)$

1

QUESTION 7: (13 marks)**Marks**

- i) Solve $\sin \theta = \frac{-\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$ **2**
- ii) (a) Sketch the graphs of $y = x^2 + x - 6$ and $y = 9 - x$ on the same number plane. Your graph should be at least $\frac{1}{3}$ page. **2**
- (b) On your graph mark the intercepts of both curves on the x-axis and y-axis, as well as the co-ordinates of their points of intersection. **2**
- (c) Hence shade the region defined by $y \geq x^2 + x - 6$ and $y \leq 9 - x$ **2**
- iii) Differentiate $y = x^3 (x + 1)^4$, expressing your answer in fully factored form **3**
- iv) Sketch the graph of $y = |x| + x$ **2**

THE END!

Year 11 Mathematics : Solutions - Semester 2 Exam 2010

i) Domain: all real x
Range: $y \leq 9$

ii) $\frac{x^2+3x+2}{x^3+8} = \frac{(x+2)(x+1)}{(x+2)(x^2-2x+4)}$
 $= \frac{x+1}{x^2-2x+4}$

iii) $5x^2+7x-12=0$
 $(5x+12)(x-1)=0$
 $x = -\frac{12}{5}, 1$

iv) a) $y = 2x^3 - 12x + 7$
 $y' = 6x^2 - 12$

b) $y = \frac{3}{x^4} = 3x^{-4}$
 $y' = -12x^{-5} = -\frac{12}{x^5}$

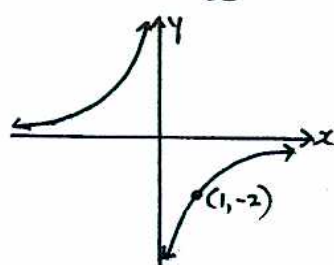
v) $\frac{a+2}{5} - \frac{a-1}{3} = \frac{3(a+2) - 5(a-1)}{15}$
 $= \frac{3a+6-5a+5}{15} = \frac{11-2a}{15}$

vi) $x^3 - x + x^2y - y = x(x^2-1) + y(x^2-1)$
 $= (x+y)(x^2-1)$
 $= (x+y)(x-1)(x+1)$

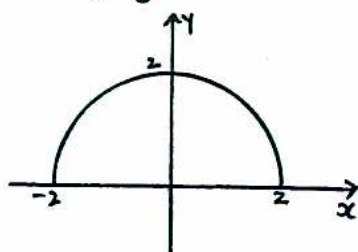
② i) $\frac{5}{2\sqrt{2}+3} \times \frac{2\sqrt{2}-3}{2\sqrt{2}-3} = \frac{5(2\sqrt{2}-3)}{8-9}$
 $= -5(2\sqrt{2}-3)$

ii) $y^{-\frac{2}{3}} = \frac{1}{\sqrt[3]{y^2}}$ or $\frac{1}{(\sqrt[3]{y})^2}$

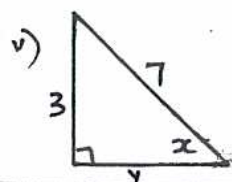
iii) a) $y = -\frac{2}{x}$



b) $y = \sqrt{4-x^2}$



iv) $y = (5-2x)^3$
 $y' = 3(5-2x)^2 \times -2$
 $= -6(5-2x)^2$



$\sin x = \frac{3}{7}$
 $y = \sqrt{49-9}$
 $= \sqrt{40}$

$\tan x < 0$
 $\therefore \text{quad } 2$
 $\cos x = -\frac{\sqrt{40}}{7}$

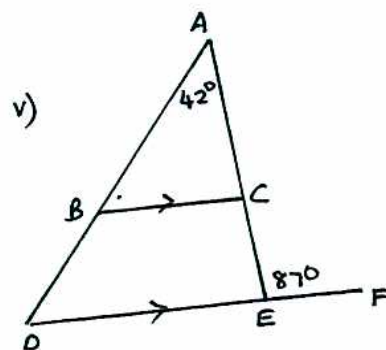
vi) $\lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2} = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+2)}$
 $= \lim_{x \rightarrow 1} \frac{1}{x+2}$
 $= \frac{1}{3}$

③ i) $f(x) = \sqrt{4x-3} = (4x-3)^{\frac{1}{2}}$
 $f'(x) = \frac{1}{2}(4x-3)^{-\frac{1}{2}} \times 4 = \frac{2}{\sqrt{4x-3}}$
 $f'(3) = \frac{2}{\sqrt{12-3}} = \frac{2}{\sqrt{9}} = \frac{2}{3}$

ii) $x-y=1 \rightarrow y=x-1$
 $xy=2$
 $x(x-1)=2$
 $x^2-x-2=0$
 $(x-2)(x+1)=0$
 $x=2, -1$
 $\therefore (x,y) = (2,1) \text{ or } (-1,-2)$

iii) $\frac{2^n \times 4^{n+1}}{8^{n-2}} = \frac{2^n \times (2^2)^{n+1}}{(2^3)^{n-2}}$
 $= \frac{2^{3n+2}}{2^{3n-6}} = 2^8$

iv) $y = \frac{x^2}{3x+1}$
 $y' = \frac{(3x+1) \times 2x - x^2 \times 3}{(3x+1)^2}$
 $= \frac{3x^2 + 2x}{(3x+1)^2}$
 $= \frac{x(3x+2)}{(3x+1)^2}$



$\angle BCE = 87^\circ$ (alt L's, $BC \parallel DF$)
 $\angle ACB = 93^\circ$ (L sum straight L)
 $\angle ABC = 45^\circ$ (L sum $\triangle ABC$)

vi) $f(x) = x^2 - 1$
 $f(a+b) = (a+b)^2 - 1 = a^2 + 2ab + b^2 - 1$
 $f(a) = a^2 - 1$
 $f(b) = b^2 - 1$
 $f(a) + f(b) = a^2 - 1 + b^2 - 1$
 $= a^2 + b^2 - 2 \neq f(a+b)$

④ i) a) $\frac{x}{\sin 70} = \frac{8.2}{\sin 30}$
 $x = \frac{8.2 \sin 70}{\sin 30}$
 $\therefore AC = 15.4 \text{ cm}$

b) Area = $\frac{1}{2} \times 15.4 \times 8.2 \sin 80$
 $= 62.2 \text{ cm}^2$

ii) $|2x-5| = 3x+1$

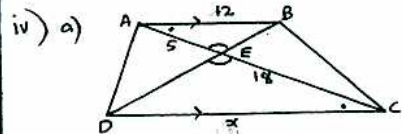
$2x-5 = 3x+1$ $2x-5 = -(3x+1)$
 $-6 = x$ $2x-5 = -3x-1$
 $5x = 4$

LHS = $|2(-6)-5| = 17$ $x = \frac{4}{5}$
RHS = $3(-6)+1 = -17$ LHS = $|2(\frac{4}{5})-5| = 3\frac{2}{5}$
 $\therefore x \neq -6$ RHS = $3(\frac{4}{5})+1 = 3\frac{2}{5}$
 $\therefore x = \frac{4}{5}$

iii) $f(x) = x^2 - 2x$

$f(x+h) = (x+h)^2 - 2(x+h)$
 $= x^2 + 2xh + h^2 - 2x - 2h$

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - (x^2 - 2x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h}$
 $= \lim_{h \rightarrow 0} 2x + h - 2$
 $= 2x - 2$

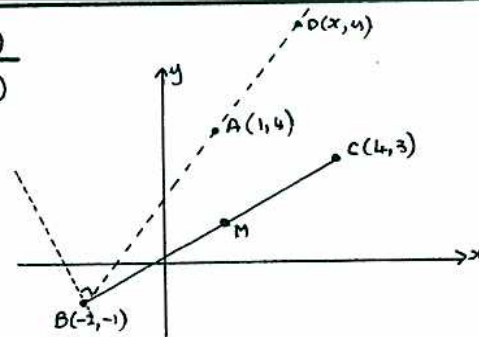


b) In $\triangle ABE$, $\triangle CDE$
 $\angle AEB = \angle CED$ (vert opp $\angle$ s)
 $\angle BAE = \angle ECD$ (alt $\angle$ s, $AB \parallel DC$)
 $\therefore \triangle ABE \parallel \triangle CDE$ (equiangular)

c) $\frac{x}{12} = \frac{18}{5}$ (ratios of corresp side)
 $\therefore x = 43.2$

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⑤



a) $BC = \sqrt{(4-(-2))^2 + (3-(-1))^2} = \sqrt{36 + 16}$
 $= \sqrt{52} = 2\sqrt{13}$

b) $M = (\frac{4-2}{2}, \frac{3-1}{2}) = (1, 1)$

c) $C(1, 1)$ $R = \sqrt{13}$
 $(x-1)^2 + (y-1)^2 = 13$

d) $m = \frac{3-1}{4-2} = \frac{2}{2} = 1$

e) eqn BC: $y - 3 = \frac{2}{3}(x - 4)$
 $3y - 9 = 2x - 8$
 $0 = 2x - 3y + 1$

f) $D = \frac{2(1) - 3(4) + 1}{\sqrt{2^2 + (-3)^2}} = \frac{9}{\sqrt{13}}$

g) $A = \frac{1}{2} \times 2\sqrt{13} \times \frac{9}{\sqrt{13}} = 9 \text{ units}^2$

h) $m_1 = -\frac{3}{2}$
 $y+1 = -\frac{3}{2}(x+2)$
 $2y+2 = -3x-6$
 $3x+2y+8=0$

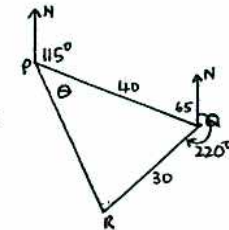
i) $B(-2, -1)$ $A(1, 4)$ $D(x, y)$
 $1 = \frac{-2+x}{2}$ $4 = \frac{-1+y}{2}$
 $2 = -2+x$ $8 = -1+y$
 $x = 4$ $y = 9$
 $\therefore D(4, 9)$

⑥

i) LHS = $\frac{1}{1-\sin x} + \frac{1}{1+\sin x}$
 $= \frac{1+\sin x + 1-\sin x}{(1-\sin x)(1+\sin x)}$
 $= \frac{2}{1-\sin^2 x} = \frac{2}{\cos^2 x}$
 $= 2 \sec^2 x = \text{RHS}$

ii) a) $\angle POR = 75^\circ$

b) $PR^2 = 40^2 + 30^2 - 2 \times 40 \times 30 \cos 75$
 $= 1878.8$
 $PR = 43.35 \text{ km}$



c) $\frac{\sin \theta}{30} = \frac{\sin 75}{43.35}$
 $\sin \theta = 0.66846$
 $\theta = 42^\circ$

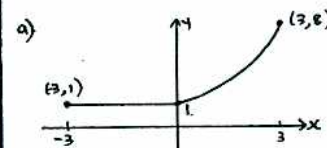
$\therefore$ Bearing is $115 + 42 = 157^\circ \text{ T}$

iii) $y = \frac{6}{\sqrt{x}} = 6x^{-1/2}$

a) $\frac{dy}{dx} = -3x^{-3/2} = \frac{-3}{\sqrt{x^3}}$

b) at $x=9$, $m = \frac{-3}{\sqrt{9^3}} = -\frac{1}{9}$

iv) $f(x) = \begin{cases} 1 & \text{for } x < 0 \\ 2^x & \text{for } x \geq 0 \end{cases}$



b) $f(-2) + f(2) = 1 + 4 = 5$

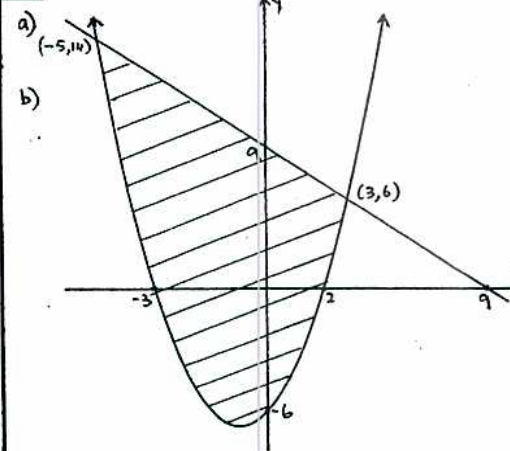
⑦ i) $\sin \theta = -\frac{\sqrt{3}}{2}$ $0^\circ \leq \theta < 360^\circ$
 $\theta = 240^\circ, 300^\circ$ Quad 3, 4 60°

ii) $y = x^2 + x - 6$
 $= (x+3)(x-2)$

$y=0$, $x=-3, 2$

$x=0$, $y=-6$

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$x^2 + x - 6 = -x + 9$

$x^2 + 2x - 15 = 0$

$(x+5)(x-3) = 0$

$x = -5, 3$

$x = -5, y = 14$ $x = 3, y = 6$

iii) $y = x^3(x+1)^4$

$\frac{dy}{dx} = x^3 \cdot 4(x+1)^3 + (x+1)^4 \cdot 3x^2$
 $= 4x^3(x+1)^3 + 3x^2(x+1)^4$
 $= x^2(x+1)^3 [4x + 3(x+1)]$
 $= x^2(x+1)^3 (7x+3)$

iv) $y = |x| + x$

